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CRYPTOCURRENCY MARKET DYNAMICS: A MACHINE LEARNING-BASED APPROACH FOR PRICE PREDICTION

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ABSTRACT

Now-a-days Crypto currencies are digital currency that can be used as long-term investments or for all kinds of transactions. Investor interest in it is high because of its fundamental values of security, transparency, and decentralization. Developing successful investing strategies requires an accurate price prediction of the crypto currency due to its unique qualities and price volatility. The price of Bitcoin, the most popular crypto currency, is predicted using this model. A multitude of factors will be considered in order to forecast the accurate prices. The change point detection technique is used to provide robust prediction performance in the unseen price range. Furthermore, unique records that exist in crypto currencies and are posted on the block chain, or "on-chain data," are gathered and used as input variables to forecast values. Experiments with real world Bit coin price data and various methods have proven the proposed system's effectiveness in Bit coin price prediction. Crypto currency price trading is a huge demand of exchange right now. It is a great advantage for investors and day-to-day traders. In this work they will use latest Machine Learning algorithms for finding out the Price Prediction of Crypto currency.These algorithms assist in identifying price prediction. These algorithms are all contrasted with one another to provide precise results.

***Keywords**: crypto currency, bit coin, block chain, On-chain-data, Point detection technique, Decentralization*

1. INTRODUCTION

Can financial, crypto, and macroeconomic shocks affect Bitcoin's price? Is the idea of an efficient market something that Bitcoin should adhere to? Does the volatility of the price of Bitcoin being influenced by other crypto currencies[1] In 2009, Bitcoin was born as the first cryptocurrency, drawing in new investors with its promise of big returns. The returns of Bitcoin, as reported by Coinbase, have increased by over 120% from 2016 to 2017, hitting USD20.000 from USD900 for purchasing of one Bitcoin token[2]. This clearly demonstrates this trend. The market valuation of Bitcoin increased dramatically from approximately USD18 billion in early 2017 to almost USD600 billion by the year's end. From its humble beginnings as a retail investment, Bitcoin has grown into a model for other digital currencies like Ethereum, XRP, and Lite coin, among many others[3]. The most valuable crypto currency in the world, bit coin is traded on more than 40 exchanges throughout the globe that accept more than 30 different currencies. Its market value is currently $9 billion USD, and more than 250,000 transactions occur there every day. Because of its youthful age and thus high price volatility much higher than that of other digital currencies Bitcoin presents a unique potential for price prediction as a currency. It differs from conventional fiat currencies in that it is open-ended; with fiat currencies, there is no comprehensive data on cash transactions or the amount of money in circulation. Long-term financial market prediction, like that of the stock market, has been studied extensively. Conventional time series prediction techniques, such the exponential smoothing models developed by Holt-Winters[4], rely on linear assumptions and work best with data that can be segmented into trend, seasonal, and noise components. When it comes to tasks like sales forecasting, where seasonal impacts are evident, this kind of methodology works better.

The Remaining Paper is organized as Follows. Section 2 Provides the Literature survey of Latest research on crypto currencies. Section 3 Provides the Proposed Methodology and Algorithm for detecting the Fake News. Section 4 provides the Data and Pre processing. Section 5 gives the results and last section concludes the work.

2. Related Works

There is a lot of uncertainty and high fluctuation in the crypto currency market, which has attracted a lot of attention recently. Investors, traders, and other

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players in the financial sector have relied heavily on accurate price predictions of crypto currencies in this context. A new line of inquiry into the reliability of crypto currency price forecasts is the application of deep learning and machine learning algorithms. Here, in survey the relevant literature, drawing attention to the most popular methods and difficulties in the field of crypto currency price forecasting.

A system utilizing LSTM and random forest regression was proposed by Chen et. al.[5]. It seeks to forecast Bitcoin's price for the upcoming day. Compared to LSTM, random forest regression offers better prediction errors (RMSE and MAPE). Bitcoin prices are influenced by various factors, including the price of ETH, oil, and US stock market indices. Since 2018, the price of ETH and the JP225 index of the Japanese stock market have been significant factors. The most accurate model for prediction is the one that uses a single lag for its explanatory variables.

In order to forecast Bit coin fluctuations, Dimitria dou et. al. [6] suggest a machine-learning architecture. Exchange rates, interest rates, and macroeconomic factors are among the twenty-four variables in the dataset. The random forest technique, logistic regression, and support vector machine are the three models that are employed. With a 66% accuracy rate, the logistic regression model performs better than the other models. The analysis recommends that the Bitcoin market reject weak form efficiency. Changing Luo et. al. [7] suggested process for recommending the timing of cryptocurrency purchases and sales. integrates deep learning, text analytics, and statistical methods. takes into account the relationship between price changes and posts on social media. considers the causal relationship between prices and social media users' mood. any transaction fees, had an average gain of 194% total. Considering fees, realized an average gain of 117% total. reliable in forecasting significant meme coin price changes. Every execution of the trading algorithm uses all of the available capital.

Huiru Xu, Binwei et. al.[8] developed a model to predict the price of Bitcoin with deep learning techniques and multiscale data. Original Bitcoin price series broken down into components with varying frequencies: high, medium, and low. utilizing deep learning techniques to improve model performance. components are reduced and intrinsic mode functions (IMFs) are rebuilt. combination of disassembled parts by means of the sample entropy technique. G. Kim, et. al.[9] The suggested system is a model for predicting bitcoin prices based on deep learning. In order to forecast Bitcoin (BTC) prices, it employs on-chain data as input variables. For consistent prediction performance, the system makes use of a change point detection technique. It makes use of the multiple long short-term memory (SAM-LSTM) self-attention-based paradigm. Experiments have shown that the technique is effective at predicting the price of bitcoin.

B. Amiri et. al.[10] suggested a model that uses sentiment analysis to forecast cryptocurrency prices in the future. Machine learning methods used to data analysis of time series. Weights depending on influence factor are applied to tweet sentiment scores. hybrid model that combines weighted sentiment scores with text analysis elements. outperformed other models in terms of cardano, EOS, and ethereum price prediction. For Bitcoin and Ripple predictions, weighted sentiment scores based on influence factor fared better than other models[11].Remaining works which are cited from[12] also discusses about Various Machine learning based methods[13-16] for prediction of Various elements which is useful for our work.

3. Methodology

Machine learning is a critical department of manufactured insights (AI). It may be divided into three categories: sponsored education, unsupervised, and reinforced learning, depending on whether a goal variable is present. In order to predict future Bitcoin prices, a relapse method using administered learning is used in this investigation. According to machine learning's cohesive execution theory, after a preset computation, a learner is formed. To obtain a high-precision learner, the learner is prepared again using training data and the review procedure is repeated. The last step is to hand over the test results to the student, who can then use them for their own work. Both the unpredictable woodland recurrence and the LSTM demonstration preparation are implemented in this paper using the open-source Python machine learning library. Arbitrary Woodland Relapse uses sky learn as its library and LSTM uses keras for research. The Pandas package handles the information collection and pre-processing.

3.1. Random Forest

One type of complex regression tree is the Random Forest. Its comprehensibility is strong, but the

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preparation tests limit the expected results. Regression tree rules state that the parent group should be divided into subgroups using a reference to a specific variable. The categorization is based on finding the least normal of all the residuals divided by squared that gather, with Condition appearing beneath.

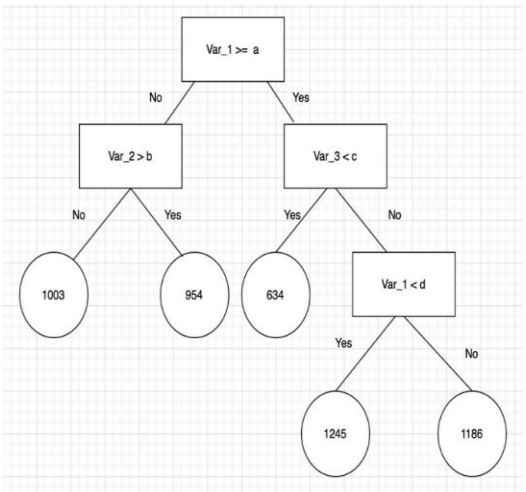

Figure1. Parameter settings

Figure 1 illustrates the parameter settings. It indicates that the random forest has 500 sub trees for regression, with an optimal depth of 10 for a single mini regression tree. The deepest point [least = 3, maximum = 20] & the greatest amount of sub-regression trees [least = 200, maximum = 1000] of the interval were tested by me. The data used for training with the error in predictions stop to fluctuate when the total amount of sub-regression trees approaches 500 or the depth limit exceeds 10, according to my subsequent testing as shown in figure 2.

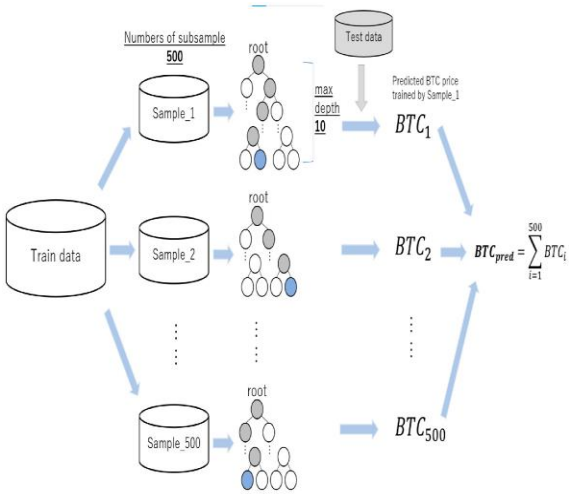

Fig 2:The random forest regression framework and parameters

The standard DNN algorithm and the RNN algorithm are not the same. When input is substituted into the model, the model's parameters are altered in addition to producing an output value. The purpose of RNN algorism is to keep the model's prior input data information. In order to compensate for the short memory issue of RNN, this work adopts the LSTM model. The three activation processes that enable data modifications regarding RNN architecture and the storage model are recall entrance, entry gate, and output gate.

This experiment's LSTM model structure centers on the property that a further level of the LSTM framework may be built by using the drop out layer that has been suggested in previous studies and substituting the model's output value. Regarding the dropout layer's parameter settings, I experimented with [less than = 10%, peak = 50%] for every layer in dropout. Additionally, the study found that the dropout value may be predicted more accurately when done in ascension order as opposed to reverse order. After balancing the risk of over learning against accuracy, the settings of every single layer of modules of [32, 64, 128, 256, 512] with the total amount of layers for each LSTM [The minimum = 2, highest = 6] are examined. As shown in figure 3Each layer's activation function is set to "ReLU," which is more effective than "sigmoid" and "tanh.". Figure. 4 below shows the LSTM structure and particular value. The last 10% of the sample set is made up of the validation information.

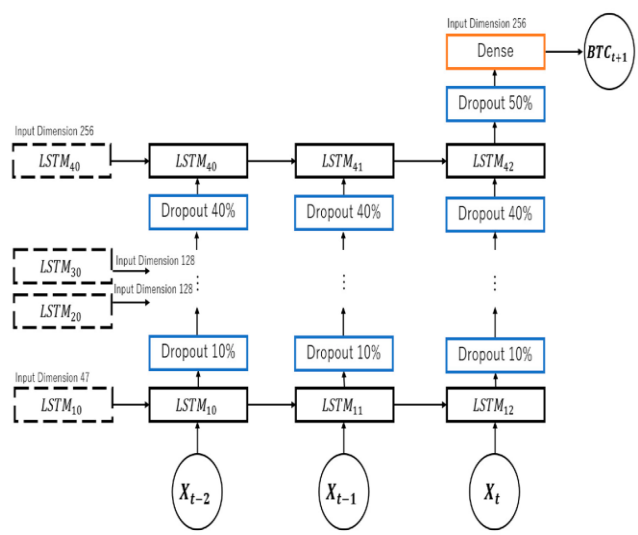

Figure 3: Parameters and framework of LSTM

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Bagging Technique & Long-Short Term Memory algorithms are employed to estimate the price of BTC on the following day with accuracy and efficiency. Compile historical Bitcoin price data with relevant criteria such as difficulty, mean block size, mining rate, location, and reasonable transfer size, and divide it into sets for testing and training. Extrapolate relevant data features that may have an impact on the price of bit coin and scale or normalize them as needed. Learn the Bagging Technique using the data set used for training, as well as the previous features and Bitcoin prices. Then, use the same training dataset to train an LSTM network. When it comes to finding sequential dependencies in time-series data, such stock prices, LSTM is quite useful. After training two models, apply the Bagging Technique to forecast using the attributes, and then use the test datasets to see how well the hybrid model works. Statistics such as Mean Absolute Percentage Error (MAPE), Mean Square Error (MSE), and others might be utilized depending on your preferences and the characteristics of your data. Finally, enter the most recent information into each of the models to predict the price of Bit coins the next day. Then, integrate the projections from each model to get the final prediction. Figure 4 depicts the overall experiment flow.

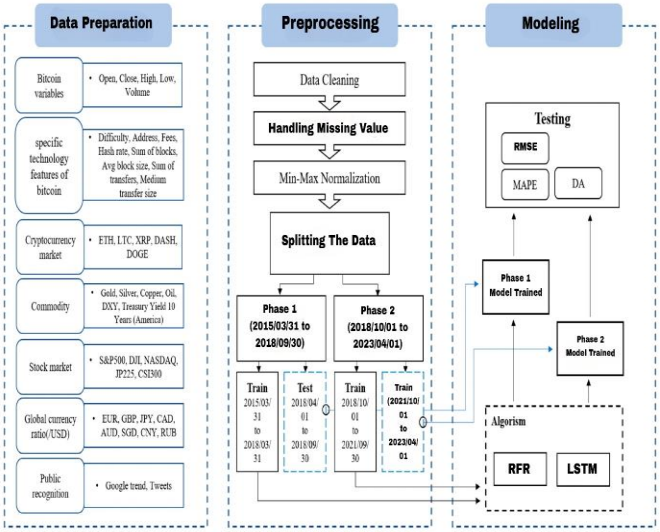

Figure 4: Proposed Methodology

4. Data and Pre processing

From March 31, 2015, to April 1, 2023, daily data were used as sample data. Bitinfocharts.com, coinmatrics.io, investing.com, Coinmarketcap.com, and Yahoo Finance were the sources of the study's data. For this experiment, the price of Bit coin in US dollars is the target variable. To forecast the Bit coin's price on the future,47 variables in all which were broken down into eight categories are employed as variables for explanation. These variables include cost variables for bit coin, technical aspects of the crypto currency, commodities, additional digital assets, foreign exchange, stock index, attention from the general public, and week's simulated variables.

4.1. Pre processing

Between March 31, 2015, and April 1, 2023, or seven natural years in total, data were gathered for the data research sample. Nonetheless, given that Bitcoin is unique in that it experienced two price bubbles at the close of 2017 and 2022, and previous research has shown that a single sample's longest duration is no more than four years. All of the data is split into two time periods based on the two parameters described above in an effort to make the model's price predictions more accurate: First, from the tail end of March 2015 all the way to the tail end of September 2018, and then, from October 1, 2018, to the beginning of October 1, 2018, there will be Phase 2. Make predictions after independently analyzing two subsamples and training models for corresponding time periods. During machine learning, training examples are used to instruct initial samples, which are then substituted with test samples for evaluation. The majority of samples (75–90%) are used for training purposes. As can be seen in Figure 5, the study's testing and training samples were precisely divided. Validation data makes up the final 10% of the original set.

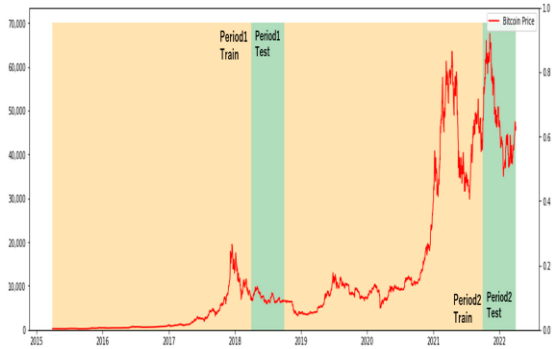

Figure 5: Trained collection and test period separation

5. Results

The test samples from Periods 1 and 2 are predicted by the trained learner, yielding the outcomes displayed in Figure 6 below. The price of Bitcoin is shown as the red line, while the price that the random forest regression learner predicted is shown as the

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green dashed line. Comparing the RMSE findings of different periods is pointless, even though Period 1's RMSE is significantly lower than Period 2's because Period 1's average Bitcoin price is also significantly lower than Period 2's. The two periods' MAPE and DA indicators are quite similar, and Period 2's forecast accuracy is marginally higher than Period 1's. Because there aren't many samples in the training intervals of Phase two with Bitcoin prices higher than USD 60,000, it's important to note that the random forest regression algorithm performs poorly in the initial stages of the taught interval of Period two when it comes to predicting the price of the crypto currency.

5.2. Results of LSTM

The LSTM algorithm is a technique for deep learning that has good prediction performance for time series data. The answer to the question of whether it is necessary to consult data from many periods in the past when projecting the price of bit coins shows that a model that just needs knowledge from earlier times yields the best prediction accuracy. As can be seen from the findings of the inaugural period (a ten-variable strategy) along with the subsequent period in Figure 7 below, the price patterns of both approaches are quite similar to the actual price. On the other hand, the smoother the forecasting curve gets plus how considerably it departs from the true cost, the more data periods are incorporated to the model.

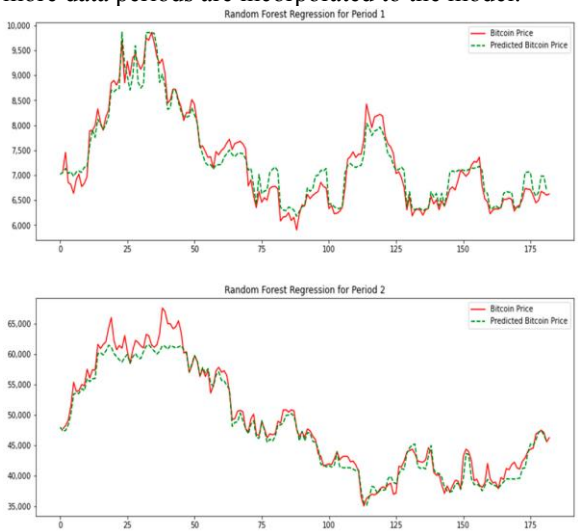

Figure 6: Based on a comparison of the real and anticipated prices using a random forest analysis

Any of the info from a particularly current time frame must be used to train framework, according to the conclusion about the number of data periods needed. This finding is in line with the theory of the efficient market. Even though when it comes to estimating the price of the digital currency, the random forest regression approach has surpassed LSTM in order by a significant margin. Despite the fact that it is not statistically superior than LSTM.

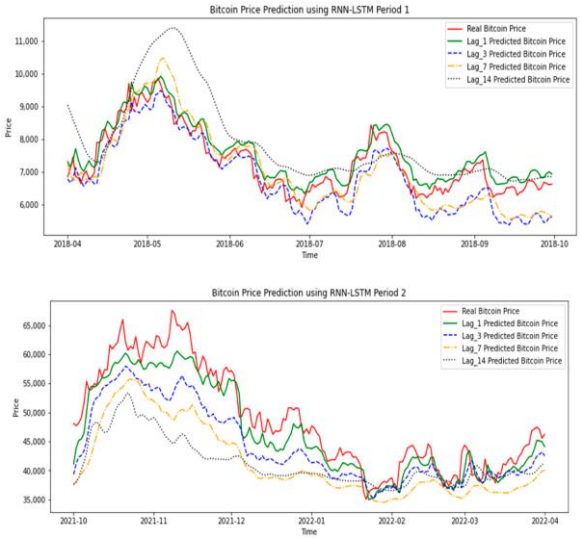

Figure 7: Precision and the number of lapses (LSTM), among in Relation.

6. Conclusion

To foresee the value of bitcoins for the upcoming moment, the following categories of several factors served as illustrative parameters in this paper: (1) influences affecting the price of Btc; (2) the characteristics of Btc technology; (3) different currency; (4) merchandise; (5) foreign currencies; (6) recognition by the public; and (7) week's false variables. It consisted of 8 groups in all, totaling 47 entries. Bagging technique operates far superior to RNN for rate analysis for forecasting. LSTM was a popular and well-recognized framework of great accuracy for finding BTC value in earlier studies. The method of machine learning termed as random forest modeling regression, it hasn't been used widely in the field by other academics previously, is used in this work to provide a result that is more accurate in predicting than LSTM layer. Despite this, one major limitation of Bagging technique that it cannot forecast outcomes that were absent from the training sets. For instance, random forest regression was unable to produce a price result higher than the previous historical high moment when the historical high for Bitcoin's price was broken. However, given the growing history of Bitcoin transactions, I believe random forest regression will outperform it once the price of Bitcoin stabilizes. A part from each model's reliability final analysis, research findings in addition revealed the model's correctness in prediction declines with an increase in the quantity of previous occurrences for the substituted explicating factors,

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regardless of whether this algorithm used is LSTM or random forest regression. The most accurate model is the one that only substitutes explaining factors from the prior era. This result is in line with the traditional efficient market theory.

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